

1. Here is a simple game. Start with a pile of raisins. A turn consists of a player taking 1, 2, 3 or 4 raisins. The last player to have a turn (that is, the player who takes the last raisin) wins. Can you find an optimal strategy? Who should win – the player who goes first or the player who goes second?
2. Tweak the previous game to allow each player to take 1 through n raisins. How has the game changed?
3. Start with two piles of raisins. A player's turn consists of taking *either* some (any number of) raisins from a single pile *or* the same number of raisins from both piles. The person to take the last raisin wins. Can you find an optimal strategy?

Raisin Games Answers and Solutions

1. It's clear that if there are 1–4 raisins, the first player can win immediately. This means that if there are 5 raisins in the pile, player one must lose. If there are 6–9 raisins in the pile, then the next player to play can leave 5 raisins and thus win. If there are 10 raisins left, the player to play is in a losing position and must leave 6–9 raisins for the next player.

This continues and we have the following conclusion: A pile with a multiple of 5 raisins is a losing position and the player to play must lose. A pile of any other size is a winning position and the player to play can always leave the next player in a losing position. The strategy is simply to take the appropriate number of raisins to leave the other player with a multiple of 5 raisins.

2. Here “multiple of 5 raisins” becomes “multiple of $n + 1$ raisins.” The rest of the answer remains the same.
3. We make a large table of positions and describe them as “Winning” or “Losing” depending on the prospects for the player to play. See Table 1.

The analysis builds from basic cases. If there is only one pile, then the player to play wins (simply take the remaining pile). If there's one raisin in each pile, then again the player to play wins (simply take 1 from each pile). If there is 1 in one pile and more than 1 raisin in the other, then the player cannot immediately win. In fact, if there are piles of 1 raisin and 2 raisins – let's call this position $(1, 2)$ – then the player to play loses (the player must leave their opponent in a winning position). But this means $(2, N)$ (with $N > 1$) and $(1, N)$ (with $N > 2$) are *winning* positions, as the player to play can leave a losing position $(1, 2)$. (Of course, (N, N) is a winning position for any $N \geq 1$, as the player to play can simply take all the raisins.) Now a simple corollary is that $(N, N + 1)$ (for $N > 1$) is a winning position, as we can leave $(1, 2)$ by taking $N - 1$ from both piles.

What if there are at least 3 raisins in each pile? If the position is $(3, 3)$, then winning is easy: take all the raisins. If the position is $(3, 4)$, then a winning strategy is to take 2 from each pile and leave $(1, 2)$. But with $(3, 5)$, the player to play must play and leave a winning position. Thus this is a losing position. But then $(3, N)$ (where $N > 5$) becomes a winning position: take raisins from the second pile to leave $(3, 5)$. Thus this is a winning position. Again a simple corollary is that $(N, N + 2)$ (for $N > 3$) is a winning position, as we can leave $(3, 5)$ by taking $N - 3$ from both piles.

If there are 4 raisins in one pile and at least that many in the other, things are now easy. From our discussions above, we've seen that $(4, 4)$, $(4, 5)$ and $(4, 6)$ are winning positions. Now $(4, 7)$ is a losing position, as we must leave a winning position. But this leads to two consequences: $(4, N)$ (where $N > 7$) is a winning position (leave $(4, 7)$ by taking $N - 7$ from the second pile) and $(N, N + 3)$ is a winning position for $N > 4$ (take $N - 4$ from both to leave $(4, 7)$).

We continue on in the table through piles of at least 8 raisins.

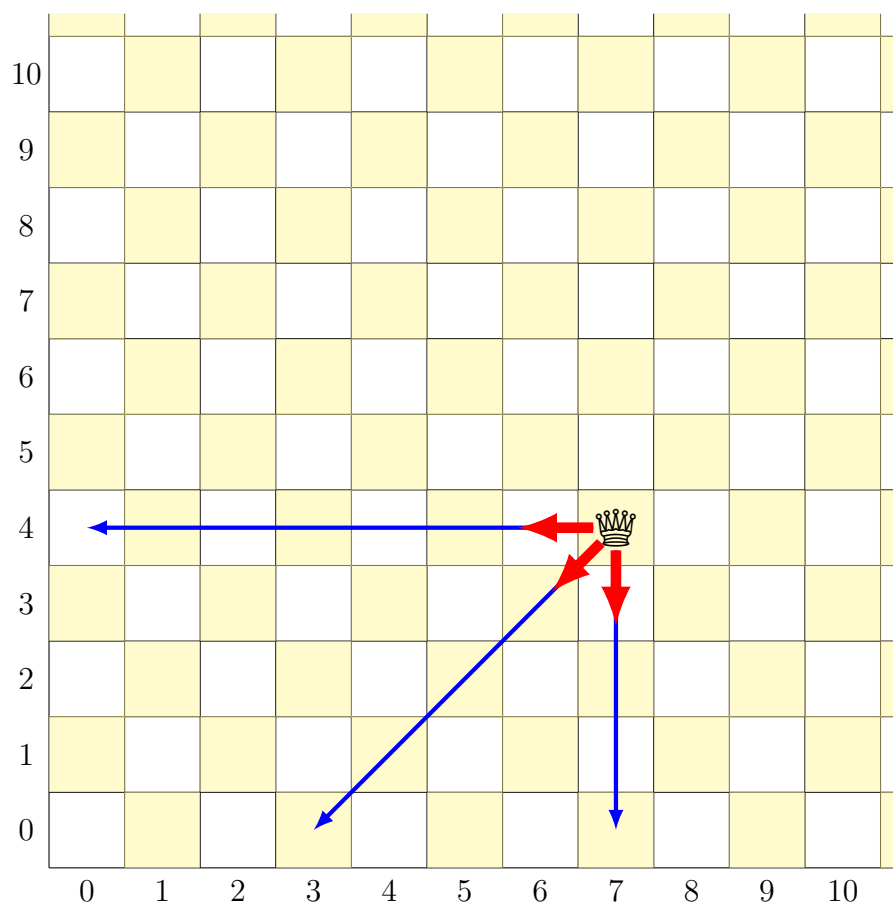
Now we might begin to look for patterns. This game is called [Wythoff's game](#) or Wythoff's Nim, as it is a variant of a very famous game called [Nim](#). (Wythoff invented it, or at least popularized it, in the early twentieth century shortly after the winning strategy for Nim was published.) Here is a paper on “simple” strategies to win the game based on Fibonacci numbers: [Simple Winning Strategies for Wythoff's Game](#) by Takemasa Ooya and Jin Akiyama.

Position	Outcome
(1, 1)	Win: take all
(1, 2)	Lose
(1, N), $N > 2$	Win: leave (1, 2)
(2, 2)	Win: take all, or leave (1, 2)
(2, N), $N > 1$	Win: take $N - 1$ to leave (1, 2)
(3, 3)	Win: take all
(3, 4)	Win: take (2, 2) to leave (1, 2)
(3, 5)	Lose
(3, N), $N > 5$	Win: take (0, $N - 5$) to leave (3, 5)
(4, 4)	Win: take all
(4, 5)	Win: take (3, 3) to leave (1, 2)
(4, 6)	Win: take (1, 1) to leave (3, 5)
(4, 7)	Lose
(4, N), $N > 7$	Win: take (0, $N - 7$) to leave (4, 7)
(5, N), $N \geq 5$	Win: take from the second pile to leave (5, 3) = (3, 5)
(6, 6)	Win: take all
(6, 7)	Win: take (2, 0) or (5, 5) to leave (4, 7) or (1, 2)
(6, 8)	Win: take (3, 3) to leave (3, 5)
(6, 9)	Win: take (2, 2) to leave (4, 7)
(6, 10)	Lose
(6, N), $N > 10$	Win: take from the second pile to leave (6, 10)
(7, N), $N \geq 7$	Win: take from the second pile to leave (7, 4) = (4, 7)
(8, 8)	Win: take all
(8, 9)	Win: take (7, 7) to leave (1, 2)
(8, 10)	Win: take (5, 5) to leave (3, 5)
(8, 11)	Win: take (4, 4) to leave (4, 7)
(8, 12)	Win: take (2, 2) to leave (6, 10)
(8, 13)	Lose
(8, N), $N > 13$	Win: take from the second pile to leave (8, 13)

Table 1: Optimal Strategy For Problem 3

(This pair also wrote a paper on the relation of the Fibonacci numbers to other, similar games: *Impact of binary and Fibonacci expansions of numbers on winning strategies for Nim-like games*, International Journal of Mathematical Education in Science and Technology, Vol. 34, pp. 121–135.)

In the literature (see, for example, Eric Duchêne and Sylvain Gravier, *Geometrical extensions of Wythoff's game*, Discrete Mathematics, Vol. 309, #11, pp. 3595–3608, available [online](#)), the game is described geometrically as moving a queen on an infinite chessboard. The squares on the board are indexed by two integers, starting at zero and increasing (so $(0, 0)$, $(5, 7)$ and so on). The position (m, n) in our description is now a square on the chessboard, and the goal is to get to the square $(0, 0)$. The allowable moves correspond to moves by a queen down and/or to the left:



The original paper by Wythoff is in *Nieuw archief voor wiskunde* on pages 199–202 of the 1905–1907 volume (available freely via [Google books](#)). He describes the k th losing position as $(\lfloor k\phi \rfloor, \lfloor k\phi^2 \rfloor)$, where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio and $\lfloor x \rfloor$ is the “floor” (or integer part) of x . (Of course these losing positions – what Wythoff calls *safe combinations*, as in they are safe to leave for your opponent – are symmetric, so if (a, b) is a losing position, so is (b, a) . We’ve implicitly assumed that $a < b$ in this description.)